

Assignment 1-INFS343

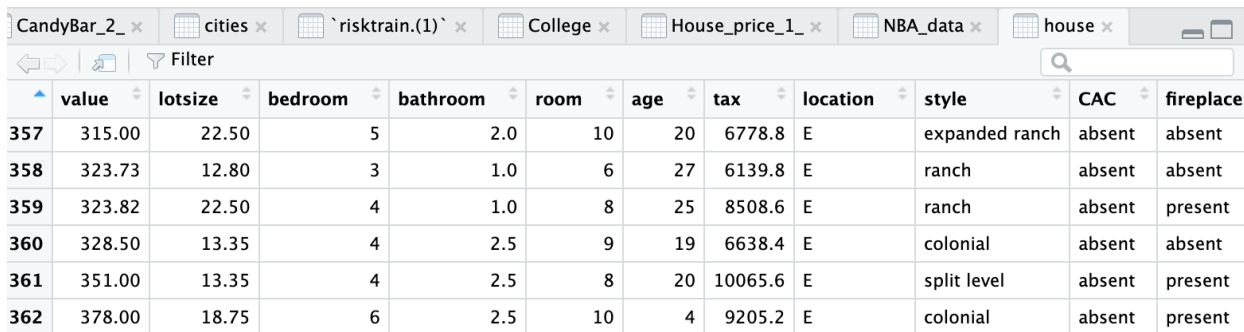
Part I: This part is on the **House.csv** data file. Please answer the following questions and provide screenshots to support your work if asked.

1. When you load the data file, we encourage you to visually examine the data set first.

1.1 How many observations and variables does this dataset contain?

Observations (rows): 362

Variables (columns): 14



	value	lotsize	bedroom	bathroom	room	age	tax	location	style	CAC	fireplace
357	315.00	22.50	5	2.0	10	20	6778.8	E	expanded ranch	absent	absent
358	323.73	12.80	3	1.0	6	27	6139.8	E	ranch	absent	absent
359	323.82	22.50	4	1.0	8	25	8508.6	E	ranch	absent	present
360	328.50	13.35	4	2.5	9	19	6638.4	E	colonial	absent	absent
361	351.00	13.35	4	2.5	8	20	10065.6	E	split level	absent	present
362	378.00	18.75	6	2.5	10	4	9205.2	E	colonial	absent	present

1.2 Are there any missing values? If so, which variables are affected?

```
> colSums(is.na(house))
  value lotsize bedroom bathroom   room   age   tax location  style   CAC
    0      0      0      0      0      0      0      0      0      0
fireplace basement   garage    EIK
    0      0      0      0
>
```

1.3 Why might missing data be a concern for analysis?

Missing data can be a problem because it may:

- Reduce the number of usable observations,
- Make results less accurate,
- Create bias in the analysis,
- Affect calculations such as averages, medians, or models.

Even though this dataset does **not** have missing values, it is still important to check before doing analysis.

- 2 Please list all the categorical variables you see in the data set. Briefly explain how you identified them.

```
> str(house)
'data.frame': 362 obs. of 14 variables:
 $ value      : num 387 486 378 445 540 ...
 $ lotsize    : num 6 10 10 8 5.18 ...
 $ bedroom    : int 2 4 3 4 4 4 3 4 3 3 ...
 $ bathroom   : num 2 2 1.5 2 3.5 2 1.5 2.5 1 2 ...
 $ room       : int 7 8 7 8 8 8 7 8 7 9 ...
 $ age        : int 30 28 30 30 15 34 30 16 26 46 ...
 $ tax        : num 5141 8100 7074 5580 8572 ...
 $ location   : chr "A" "B" "B" "A" ...
 $ style      : chr "ranch" "ranch" "split level" "cape" ...
 $ CAC        : chr "present" "present" "present" "present" ...
 $ fireplace  : chr "absent" "absent" "absent" "present" ...
 $ basement   : chr "yes" "yes" "yes" "yes" ...
 $ garage     : chr "one-car garage" "two-car garage" "one-car garage" "one-car garage" ...
 $ EIK        : chr "present" "present" "present" "present" ...
```

Location, Style, CAC, Fireplace, Basement, Garage, EIK

I identified the categorical variables using the `str(house)` output. Variables labeled as "chr" (character) are categorical because they represent **groups or categories instead of numeric values**.

- 3 How to calculate the mean or median of one variable? Please write down the R code for the mean and median for house value and house age. Remember to write down both codes and the results you get.

	R code for mean/Value	R code for median/Value
House value	<code>mean(house\$value) = 320.1335</code>	<code>median(house\$value) = 315</code>
House age	<code>mean(house\$age) = 33.42541</code>	<code>median(house\$age) = 33.5</code>

- 4 Please write down the R code and the results for the house tax and house lotsize.

	R code for minimum/Value	R code for maximum/Value
House tax	<code>min(house\$tax) = 1800</code>	<code>max(house\$tax) = 11646</code>
House lotsize	<code>min(house\$lotsize) = 3.55</code>	<code>max(house\$lotsize) = 37.5</code>

- 5 Please use summary function on the data frame and provide a screenshot of the results. Make sure you include all the variables. Based on the summary output, identify one variable with high variability and explain why.

```
> summary(house)
```

value	lotsize	bedroom	bathroom	room	age
Min. :180.0	Min. : 3.550	Min. :1.000	Min. :1.00	Min. : 4.000	Min. : 1.00
1st Qu.:279.0	1st Qu.: 6.000	1st Qu.:3.000	1st Qu.:1.00	1st Qu.: 6.000	1st Qu.:28.00
Median :315.0	Median : 7.500	Median :3.000	Median :1.50	Median : 7.000	Median :33.50
Mean :320.1	Mean : 9.327	Mean :3.575	Mean :1.61	Mean : 7.052	Mean :33.43
3rd Qu.:342.0	3rd Qu.:10.000	3rd Qu.:4.000	3rd Qu.:2.00	3rd Qu.: 8.000	3rd Qu.:37.00
Max. :558.0	Max. :37.500	Max. :7.000	Max. :3.50	Max. :12.000	Max. :95.00

tax	location	style	CAC	fireplace
Min. : 1800	Length:362	Length:362	Length:362	Length:362
1st Qu.: 5040	Class :character	Class :character	Class :character	Class :character
Median : 5940	Mode :character	Mode :character	Mode :character	Mode :character
Mean : 6104				
3rd Qu.: 7148				
Max. :11646				

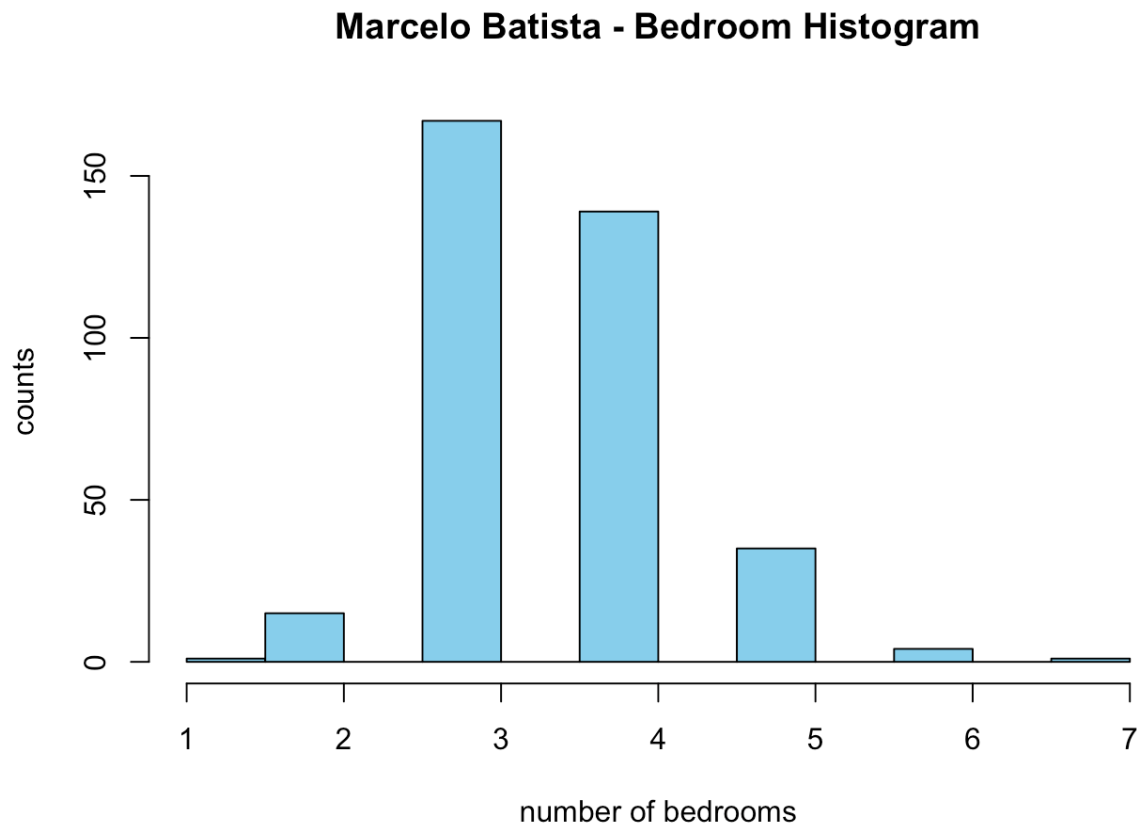
basement	garage	EIK
Length:362	Length:362	Length:362
Class :character	Class :character	Class :character
Mode :character	Mode :character	Mode :character

Based on the summary output, lotsize appears to have high variability because its values are spread out over a wide range. The minimum lotsize is 3.55, while the maximum is 37.5, which shows a large difference between smaller and larger properties. This suggests that house lot sizes vary a lot across the dataset.

One variable with high variability is house tax because the values range from 1800 to 11646. This shows that taxes differ a lot between houses in the dataset.

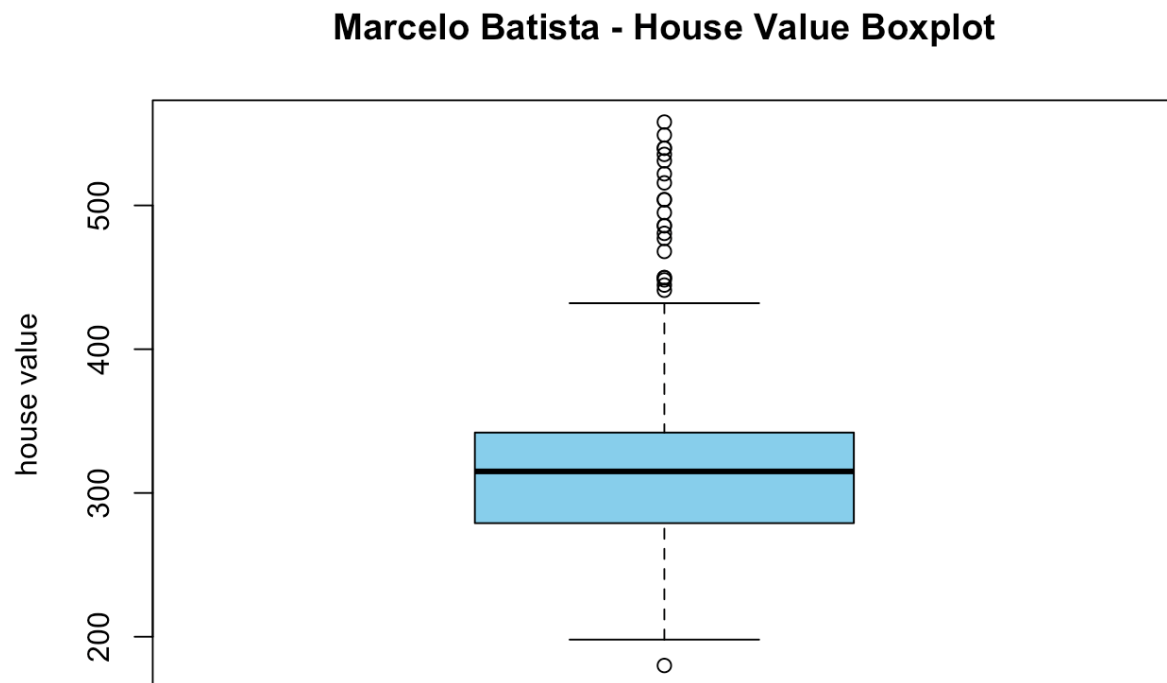
- 6 Please use histogram function to make a histogram graph for the bedroom. Make sure that you have “*You name*” (please spell out your full name here) in the main title for the graph. Change the x-axis to “number of bedrooms” and the y-axis to “counts”. Remember to change this histogram graph to your favorite color. Screenshot the code and the histogram graph in order to receive full credits.

```
> hist(house$bedroom,  
+      main = "Marcelo Batista - Bedroom Histogram",  
+      xlab = "number of bedrooms",  
+      ylab = "counts",  
+      col = "skyblue",  
+      border = "black")
```



- 7 Please use boxplot function to make a graph for the house value. Make sure that you have “*Your name*” (please spell out your full name here) in the main title for the graph. Change the y-axis to “house value”. Remember to change this graph to your favorite color. Screenshot the code and the boxplot graph. Do you see any outliers for house values? If so, approximately what values do they occur at?

```
> boxplot(house$value,  
+         main = "Marcelo Batista - House Value Boxplot",  
+         ylab = "house value",  
+         col = "skyblue",  
+         border = "black")  
> |
```



Yes, the boxplot shows outliers for house value. They appear mostly above 440, with the highest values near 558, and there is also a possible low outlier around 180.

Part II: We are using *NBA_data.xlsx* file for the second part.

With player salaries continuing to rise in professional sports, understanding what factors contribute to player compensation has become increasingly important. The data file used here contains information on NBA players, including their annual salary (Salary in \$), age (Age in years), average minutes played per game (Minutes_per_game), average points scored per game (Points), average assists per game (Assists), and personal_fouls. These variables allow us to examine how both individual performance and playing time are associated with player earnings, as well as whether different aspects of performance interact to influence salary.

1. Estimate a regression model (Model 1) for the Salary, using five variables: age, minutes per game, points and assists. Provide screenshots of the codes and the output.

Salary=b0+b1(age)+b2(minutes_per_game)+b3(points)+b4(assists) + b5(personal_fouls)

```
> model1 <- lm(salary ~ age + minutes_per_game + points + assists + personal_fouls, data = NBA_data)
> summary(model1)
```

Call:
lm(formula = salary ~ age + minutes_per_game + points + assists +
 personal_fouls, data = NBA_data)

Residuals:

Min	1Q	Median	3Q	Max
-10642990	-2138594	-236318	1631522	14533512

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-1679328	1182169	-1.421	0.156
age	-29402	44941	-0.654	0.513
minutes_per_game	47526	66390	0.716	0.474
points	611271	86857	7.038	7.56e-12 ***
assists	3048	151477	0.020	0.984
personal_fouls	378083	422525	0.895	0.371

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3861000 on 439 degrees of freedom
Multiple R-squared: 0.4621, Adjusted R-squared: 0.456
F-statistic: 75.42 on 5 and 439 DF, p-value: < 2.2e-16

1.1. Write out the complete regression equation with beta coefficients.

The estimated regression equation for Model 1 is:

Salary = -1,679,328 - 29,402(age) + 47,526(minutes_per_game) + 611,271(points) + 3,048(assists) + 378,083(personal_fouls)

1.2. Which variable has the strongest association with salary? Explain why.

The variable with the strongest association with salary is points. Since it has the largest coefficient among the important variables (611,271) and is also highly statistically significant with a p-value of 7.56×10^{-12} . This means that players who score more points earn much higher salaries.

2. Estimate a regression model (Model 2) for the Salary, using four variables: age, minutes per game, points and assists, plus the interaction between points and assists. Provide screenshots of the codes and the output.

```
> model2 <- lm(salary ~ age + minutes_per_game + points + assists + points:assists, data = NBA_data)
> summary(model2)
```

Call:

```
lm(formula = salary ~ age + minutes_per_game + points + assists +
    points:assists, data = NBA_data)
```

Residuals:

Min	1Q	Median	3Q	Max
-11557019	-2229087	-128665	1374659	13724798

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-954354	1163524	-0.820	0.412532
age	-24471	44053	-0.555	0.578840
minutes_per_game	221770	63831	3.474	0.000563 ***
points	213170	125350	1.701	0.089727 .
assists	-1312487	334386	-3.925	0.000101 ***
points:assists	96215	22924	4.197	3.27e-05 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3790000 on 439 degrees of freedom

Multiple R-squared: 0.4819, Adjusted R-squared: 0.476

F-statistic: 81.66 on 5 and 439 DF, p-value: < 2.2×10^{-16}

2.1. Write out the complete regression equation with beta coefficients.

The estimated regression equation for Model 2 is:

Salary = -954,354 - 24,471(age) + 221,770(minutes_per_game) + 213,170(points) - 1,312,487(assists) + 96,215(points × assists)

2.2. Which variable has the strongest association with salary? Explain why.

In Model 2, the variable with the strongest association with salary is the interaction between points and assists (points × assists). Since it is highly statistically significant with a p-value of 3.27×10^{-5} , and it shows that the effect of scoring on salary depends on the number of assists a player has. In other words, players who both score and assist more earn higher salaries.

3. Compare the Two Models

3.1. Which model fits the data better? (Use R-squared or adjusted R-squared)

Model 2 fits the data better because it has a higher R-squared and adjusted R-squared than Model 1. Model 1 has R-squared = 0.4621 and adjusted R-squared = 0.4560, while Model 2 has R-squared = 0.4819 and adjusted R-squared = 0.4760. Both values are higher in Model 2, it explains more of the variation in salary.

3.2. What does the inclusion of the interaction term change in interpretation?

The interaction term changes the interpretation because it means the effect of points on salary depends on the number of assists, and the effect of assists also depends on points. In Model 2, scoring and assisting are not treated as completely separate effects. Instead, the model suggests that players who are strong in both scoring and assisting tend to earn more than we would expect from either variable alone.

4. Using Model 2, predict salary for a player with:

salary = $-954,354 - 24,471(25) + 221,770(30) + 213,170(\text{points}) - 1,312,487(5) + 96,215(\text{points} \times 5)$

- Age = 25
- Minutes per game = 30
- Assists = 5

Compute predicted salary for:

4.1. Points = 10

salary = $-954,354 - 24,471(25) + 221,770(30) + 213,170(10) - 1,312,487(5) + 96,215(10 \times 5)$
= 5,466,986

4.2. Points = 15

salary = $-954,354 - 24,471(25) + 221,770(30) + 213,170(15) - 1,312,487(5) + 96,215(15 \times 5)$
= 8,938,211

4.3. Points = 20

salary = $-954,354 - 24,471(25) + 221,770(30) + 213,170(20) - 1,312,487(5) + 96,215(20 \times 5)$
= 12,409,436